

DBA5101 – Analytics in Managerial Economics

Homework 1

This homework is worth 50 marks, and you have been awarded 10 bonus marks in advance. It is due on **September 2, 2025, 23:59**. Late submissions will not be accepted. Please submit your answers as a single PDF file through Canvas. If you submit after **August 31, 2025, 23:59**, the 10 bonus marks will be deducted.

1. [Competitive market] Suppose that the total cost of producing T-shirts is given by $c(q) = 2q + 2q^2$, where q is the quantity supplied by the firm.
 - (a) Show that the competitive supply behavior of the firm is given by $q = \frac{P}{4} - \frac{1}{2}$, where P is the market price of a T-shirt.
 - (b) If there are 100 such firms in the industry, each acting as a perfect competitor, show that the market supply curve is, in inverse form, given by: $P = 2 + \frac{Q}{25}$, where Q is the aggregate quantity of the market.
2. [Monopoly] Consider a firm whose total cost is $c(q) = c_v(q) + F$ where $F > 0$ denotes the fixed costs. We assume that $c_v(q) = m \cdot q$, where $m \geq 0$ is a fixed number.
 - (a) Show that the average cost is always decreasing, and the marginal cost is always below average cost.
 - (b) Suppose a leading firm entered the industry one year ago and it has been earning high profits (greater than F). Suppose there has not been technology improvement in this industry during the last year. Now, another firm is considering entering the industry. Would the new firm survive the competition? Explain briefly.
 - (c) My teaching assistant tells me that Microsoft (a software company in US) is a leading firm and has high operational leverage. What does she mean with this?
 - (d) It seems that there are several firms competing with Microsoft. However, our theory predicts that this industry is likely a monopoly. Why is the reality different from our theory? Please provide at least two reasons.